IoT Project 10 - Smart Home Energy Optimization Assistant

Smart Home Energy Optimization Assistant

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1. Introduction

The Smart Home Energy Optimization Assistant is an innovative, privacy-focused system designed to help households monitor, predict, and intelligently manage their energy consumption. Many households still lack actionable insights into how and when energy is consumed by individual appliances. The system is designed to monitor appliance-level energy consumption using smart plugs, analyze usage patterns, and forecast future energy demand.

The project is currently implemented as a complete simulation. The system simulates smart home energy consumption, processes data in real time, stores historical records, performs forecasting, and visualizes insights through an interactive dashboard. Each component is containerized to ensure scalability, portability, and ease of deployment.

The key objectives of the project are to simulate appliance-level energy monitoring using virtual smart plugs, to store and process energy consumption data using cloud services, to analyze and forecast energy usage patterns, to suggest optimized appliance usage schedules, and to demonstrate the feasibility of smart energy optimization systems.

2. System Architecture

2.1 ESP32 ECU Simulation using Wokwi

Wokwi [1] is used to simulate ESP32-based Electronic Control Units that represent the smart plugs without using physical hardware. Potentiometers are used as virtual inputs to simulate varying appliance power consumption. The simulation generates energy usage data in real time.

The platform utilizes the Wokwi ESP32 Simulator running MicroPython code [2]. The hardware components consist of two sliding potentiometers connected to GPIO pins 34 and 35, which serve as interactive controls for power consumption simulation. The system establishes a TLS-enabled connection to HiveMQ Cloud [3] using port 8883 for secure MQTT communication.

Telemetry data is published every 5 seconds to the topic pattern `tele/sonoff_pow_XX/SENSOR`, where XX represents the device identifier.

The system simulates a total of seven devices, with two devices controlled via hardware potentiometers and five devices generating software-simulated data. The data format follows a JSON structure containing voltage measurements ranging from 220 to 235 volts, current readings from 0 to 2.27 amperes, power consumption values from 0 to 500 watts, cumulative energy consumption in kilowatt-hours, unique device identifiers, timestamps, and environmental data including temperature and humidity readings.

2.2 Simulated Smart Plugs using Sonoff

Initially, the project planned to include Sonoff [4] devices as additional simulated smart plugs within the system. However, this component was replaced by the Wokwi-based simulation approach for consistency and better hardware emulation. The current implementation focuses entirely on the Wokwi ESP32 simulation, which provides a more unified and realistic representation of smart plug behavior.

2.3 Data Flow and Logic Engine using Node-RED Dashboard

Node-RED [5] acts as the core logic and data orchestration engine for the entire system. It receives data from the simulated devices, applies business logic and rule-based decisions on the inputs, and forwards data to the database as well as the forecasting services.

The system implements three primary flows within Node-RED. The first flow handles telemetry processing, where an MQTT subscriber node listens to the topic pattern `tele/+/SENSOR` to receive data from all devices simultaneously. A function node parses the incoming JSON payloads, extracts power values and device identifiers, and formats the data appropriately for InfluxDB storage. The processed data is then written to the `energy_usage` measurement with appropriate tags and fields.

The second flow implements predictive control through Edge AI logic. This flow periodically queries the latest forecasts from the `energy_forecast` measurement in InfluxDB. A function node evaluates the forecast data against a predefined threshold of 400 watts. When the system predicts that a device will exceed this threshold within the forecast window, it automatically sends an OFF command via MQTT to the appropriate device topic before the overload occurs. Simultaneously, the system logs alert information to the `system_alerts` measurement for tracking and analysis.

The third flow provides manual control capabilities through HTTP endpoints. These endpoints accept REST API requests from the Grafana dashboard, allowing users to manually send ON or OFF commands to specific devices. The function node validates the incoming requests, formats

the appropriate MQTT messages, and publishes them to the command topics for device control.

2.4 Time-Series Database using InfluxDB

InfluxDB [6] serves as the time-series database for storing energy consumption data, including appliance-level power usage, timestamps, and device identifiers, as well as historical metrics. The database is optimized for high-frequency sensor data writing and reading operations.

The database structure consists of a default bucket named `energy_data` that contains three primary measurements. The `energy_usage` measurement stores real-time consumption data from all devices, including power, voltage, current, energy consumption, temperature, and humidity values. The `energy_forecast` measurement contains AI prediction results with confidence intervals, including predicted values and upper and lower bounds for each forecast timestamp. The `system_alerts` measurement maintains a history of alert events and control actions, including device identifiers, alert types, predicted values that triggered alerts, and actions taken.

The system uses Flux as the query language for advanced time-series queries, enabling complex aggregations, windowing operations, and data transformations. Data retention policies are configurable, allowing the system to manage storage requirements while maintaining historical data for analysis and forecasting purposes.

2.5 Data Visualization using Grafana

Grafana [7] provides interactive dashboards for real-time energy monitoring, historical usage analysis, comparative appliance consumption insights, and forecasted energy trends. These visualizations help users understand energy patterns and make informed decisions about energy consumption.

The dashboard features include real-time power consumption gauges that display current consumption values for each device, historical consumption time-series graphs that show consumption patterns over time, AI forecast visualizations with upper and lower confidence bounds that illustrate predicted future consumption, manual control buttons that allow users to send ON or OFF commands directly from the dashboard, alert history displays that show when and why automatic actions were taken, and comparative analysis panels that enable users to compare consumption patterns across multiple devices simultaneously.

The dashboard connects to InfluxDB as its data source, using Flux queries to retrieve and aggregate data efficiently. The visualization panels update in real-time, typically refreshing every 5 seconds to provide current information to users.

2.6 Forecasting Services using Prophet

Facebook Prophet [8] serves as the forecasting engine to predict future energy consumption based on data retrieved from InfluxDB. The service outputs optimized appliance usage schedules and provides insights into future consumption patterns.

The forecasting process begins by extracting the last 24 hours of historical data for each device from InfluxDB. The Prophet model is then trained on these power consumption patterns, automatically detecting daily and weekly seasonality components. Once trained, the model generates 6-hour ahead forecasts with confidence intervals, providing both point predictions and uncertainty estimates. These forecasts are stored back to InfluxDB in the `energy_forecast` measurement for visualization and use by the Edge AI control logic.

The forecasting service runs periodically on a configurable schedule, typically executing hourly to provide updated predictions as new data becomes available. Each device receives its own independent forecast, allowing the system to understand and predict the unique consumption patterns of individual appliances.

2.7 Docker-based Backend Infrastructure

Docker [9] is used to containerize and manage all backend services, including Node-RED, InfluxDB, Grafana, and the Prophet forecasting service. This containerization enables isolated and reproducible environments, easy service orchestration, and simplified deployment and maintenance.

The containerized services include InfluxDB running on port 8086 with persistent volumes for data storage, Node-RED running on port 1880 with flows persisted in volumes, Grafana running on port 3000 with dashboard configurations persisted, and a custom Python container for the forecast service with Prophet dependencies and scheduled execution capabilities.

All services operate on the same Docker network, enabling internal service discovery through service names while maintaining port mapping for external access. Persistent volumes ensure that data, configurations, and flows are maintained across container restarts, providing reliability and data persistence.

3. Edge/AI Methodology

3.1 Edge Computing Architecture

The system implements a hybrid edge-cloud architecture where intelligent decision-making occurs at the edge layer through Node-RED, while heavy computational tasks such as forecasting are performed by dedicated services. This approach minimizes latency for critical control decisions while leveraging cloud resources for complex analytics.

The edge layer components provide real-time processing capabilities, with Node-RED processing incoming MQTT telemetry with sub-second latency. Rule-based control mechanisms

enable immediate response to threshold violations, particularly the 400-watt limit established for device protection. The predictive control system implements Edge AI logic that evaluates forecast data and triggers preventive actions before overloads occur, rather than simply reacting to current conditions. Local decision-making ensures that control commands are sent directly to devices without requiring cloud round-trip communication, reducing response times and improving system reliability.

3.2 Artificial Intelligence Methodology

3.2.1 Time-Series Forecasting with Prophet

Facebook Prophet [8] was chosen for its robustness in handling time-series data with automatic seasonality detection for daily and weekly patterns, missing data tolerance, outlier resilience, and interpretable components including trend, seasonality, and holiday effects. The methodology follows the approach described by Taylor and Letham [10], which provides a scalable framework for forecasting at scale.

The training process begins with data collection, where the system retrieves the last 24 hours of power consumption data per device from InfluxDB. Data preprocessing handles missing values, validates data quality, and formats timestamps appropriately for Prophet, using datetime values for the `ds` column and power values for the `y` column. Model training configures Prophet with default hyperparameters, fits the model to historical consumption patterns, and captures device-specific usage behaviors. Forecast generation creates a future dataframe for the next 6 hours, generates predictions with confidence intervals, and accounts for uncertainty in predictions through upper and lower bounds.

The forecast output provides a 6-hour prediction horizon with timestamp, predicted power value, lower bound, and upper bound for each forecast point. The system updates forecasts on a configurable schedule, typically hourly, and maintains separate Prophet instances for each smart plug to capture individual device characteristics.

3.2.2 Edge AI Control Logic

The Edge AI component implements predictive protection rather than reactive threshold monitoring. The algorithm begins with forecast evaluation, where Node-RED queries the latest forecasts from InfluxDB. Threshold analysis checks if any predicted value exceeds the 400-watt threshold within the forecast window. If the forecast indicates an overload, the system automatically sends an OFF command via MQTT before the event occurs, preventing the overload rather than reacting to it. Alert logging records the event with timestamp, device identifier, predicted value, and action taken for analysis and tracking.

The advantages of this approach include proactive protection that prevents overloads rather than reacting to them, multi-device coordination that evaluates all devices independently,

reduced false positives through the use of forecast confidence intervals to avoid unnecessary shutdowns, and user override capabilities that ensure manual control is always available through the Grafana interface.

3.3 Integration of Edge and Cloud Components

The data flow begins with Wokwi ESP32 devices publishing telemetry data to the MQTT broker, which is then received by Node-RED for edge processing. Node-RED forwards the data to InfluxDB for cloud storage, where it becomes available for the Prophet forecasting service. The forecasting service queries historical data, generates predictions, and stores forecasts back to InfluxDB. Node-RED then queries these forecasts for Edge AI control decisions, and when necessary, publishes control commands back through the MQTT broker to the devices.

This architecture provides a clear separation of concerns between the edge layer, which handles low-latency control decisions and real-time data processing, and the cloud layer, which manages historical storage, complex analytics, and long-term forecasting. The hybrid approach combines the responsiveness of edge computing with the computational power of cloud resources, creating an efficient and scalable system architecture.

4. Implementation Details

4.1 Hardware Simulation (Wokwi)

The ESP32 MicroPython code structure begins with WiFi configuration, connecting to the Wokwi-GUEST network. The MQTT client utilizes the `umqtt.robust` library with TLS support for secure communication. ADC reading functionality captures potentiometer values from GPIO pins 34 and 35, converting analog signals to digital values. For hardware devices, power is calculated from potentiometer ADC values, mapping the 0-4095 ADC range to a 0-500 watt power range. Simulated devices generate random power values with smooth transitions to avoid abrupt changes that would not occur in real-world scenarios.

The JSON payload structure matches the Sonoff POW R2 format, ensuring compatibility with existing smart home ecosystems. Key implementation features include TLS/SSL encryption for secure MQTT communication, Home Assistant auto-discovery configuration for automatic device recognition, Last Will and Testament functionality for device availability monitoring, robust error handling and reconnection logic for network reliability, and timestamp generation compatible with InfluxDB requirements.

4.2 MQTT Communication

The broker configuration utilizes HiveMQ Cloud [3] as the MQTT provider, using MQTT over TLS on port 8883 for secure communication. Authentication is handled through username and

password credentials, and the system uses QoS level 1 for reliable message delivery with at-least-once semantics.

The topic structure follows a hierarchical pattern where `tele/{device_id}/SENSOR` carries telemetry data, `tele/{device_id}/LWT` indicates device availability status, `cmnd/{device_id}/POWER` receives command messages for ON/OFF control, `stat/{device_id}/POWER` confirms status changes, and `homeassistant/sensor/{device_id}_power/config` provides auto-discovery configuration for Home Assistant integration.

4.3 Node-RED Flow Implementation

The telemetry processing flow begins with an MQTT input node that subscribes to the `tele/+/SENSOR` topic pattern using a wildcard to receive messages from all devices. A function node parses the incoming JSON payloads, extracts the device identifier from the topic name, and formats the data appropriately for InfluxDB storage. The InfluxDB output node writes the processed data to the `energy_usage` measurement with device_id as a tag and power, voltage, current, and energy_kwh as fields.

The predictive control flow implements Edge AI functionality by querying the latest forecasts from the `energy_forecast` measurement using an InfluxDB input node. A function node implements the threshold check logic, identifying devices that are forecasted to exceed the 400-watt limit. When such devices are identified, an MQTT output node publishes an OFF command to the appropriate `cmnd/{device_id}/POWER` topic. Simultaneously, an InfluxDB output node logs the alert to the `system_alerts` measurement for historical tracking.

The manual control flow provides REST API endpoints through HTTP input nodes that accept ON/OFF command requests. A function node validates the incoming requests, formats the appropriate MQTT messages, and an MQTT output node sends the commands to the appropriate device topic for execution.

4.4 InfluxDB Schema Design

The `energy_usage` measurement structure includes device_id as a tag for efficient querying and filtering, with fields for power in watts, voltage in volts, current in amperes, energy consumption in kilowatt-hours, temperature in Celsius, and humidity as a percentage. Timestamps are automatically set by InfluxDB upon data insertion.

The `energy_forecast` measurement includes device_id as a tag and fields for yhat representing the predicted power in watts, yhat_lower representing the lower confidence bound, and yhat_upper representing the upper confidence bound. The timestamp represents the forecast timestamp, which is a future time point.

The `system_alerts` measurement includes device_id and alert_type as tags, with fields for predicted_power representing the predicted value that triggered the alert, action_taken indicating the action performed, and alert_active as an integer flag. The timestamp indicates when the alert occurred.

4.5 Prophet Forecasting Service

The Python implementation utilizes dependencies including pandas for data manipulation, Prophet [8] for time-series forecasting, and influxdb-client for database interaction. Data retrieval uses the InfluxDB Python client with Flux queries to extract historical consumption data. Model training creates a Prophet instance and fits it to the historical data, automatically detecting seasonality patterns. Forecast generation creates a future dataframe, generates predictions with confidence intervals, and stores the results back to InfluxDB.

The service runs on a configurable schedule, typically executing hourly. It processes each device independently, allowing for device-specific pattern recognition. Error handling is implemented gracefully with comprehensive logging to ensure system reliability and debugging capabilities.

4.6 Docker Compose Configuration

The services configuration includes the official InfluxDB v2.7 image running on port 8086 with persistent volumes for data storage, the official Node-RED image running on port 1880 with flows persisted in volumes, the official Grafana image running on port 3000 with dashboard configurations persisted, and a custom Python image for the forecast service with Prophet dependencies and scheduled execution capabilities.

Networking configuration places all services on the same Docker network [9], enabling internal service discovery through service names while maintaining port mapping for external access. Volume configuration ensures data persistence for InfluxDB data, Node-RED flows backup, and Grafana dashboard configurations, maintaining system state across container restarts.

4.7 Grafana Dashboard Configuration

The data source configuration connects Grafana [7] to InfluxDB using the Flux query language, with authentication handled through API tokens. The dashboard panels include gauge panels for real-time power consumption per device, time series panels for historical consumption and forecasts with confidence bands, stat panels for current values and summaries, table panels for alert history, and button panels for manual control interface with HTTP requests to Node-RED endpoints.

5. Dataset/Measurements

5.1 Data Collection Methodology

The simulation parameters include a total of seven smart plugs, with two devices controlled via hardware potentiometers and five devices generating software-simulated data. The system operates with a sampling rate of 5-second intervals, providing continuous streaming of data points during simulation runtime. The duration of data collection is configurable, typically running for hours to days during testing phases to generate sufficient historical data for forecasting and analysis.

5.2 Data Characteristics

Power consumption values range from a minimum of 0 watts when devices are off to a maximum of 500 watts, representing the simulated appliance limit. Hardware-controlled devices map potentiometer positions from 0 to 100 percent to power values from 0 to 500 watts, providing interactive control over consumption simulation. Simulated devices generate random values with smoothing algorithms to avoid abrupt changes that would not occur in real-world scenarios.

Voltage measurements typically range from 220 to 235 volts, following European standard electrical specifications. The system simulates realistic voltage fluctuations that occur in actual electrical systems. Additional metrics include current values calculated from power and voltage relationships, ranging from 0 to 2.27 amperes, cumulative energy consumption tracked in kilowatt-hours, simulated ambient temperature values ranging from 20 to 30 degrees Celsius, and simulated humidity levels ranging from 30 to 70 percent.

5.3 Data Format

The MQTT JSON payload structure includes voltage measurements in volts, current readings in amperes, cumulative energy consumption in kilowatt-hours, appliance names for identification, unique device identifiers, timestamps for temporal tracking, temperature readings in Celsius, humidity values as percentages, and power consumption in watts. This structure ensures compatibility with various smart home platforms and provides comprehensive device information.

The InfluxDB line protocol format follows a specific structure with measurement names, tags for indexing and filtering, fields for actual data values, and timestamps for temporal organization. This format enables efficient storage and querying of time-series data while maintaining data integrity and enabling complex analytical operations.

5.4 Data Quality and Validation

Validation rules ensure data integrity by checking that power values remain within the 0 to 500 watt range, voltage values stay within the 220 to 235 volt range, timestamps are properly

formatted and validated, device identifiers remain consistent across messages, and JSON schema validation ensures proper data structure.

Error handling mechanisms filter invalid data at the Node-RED processing level, handle missing values through Prophet interpolation capabilities, and implement connection failure logging with automatic retry mechanisms to ensure system reliability and data continuity.

5.5 Dataset Size and Storage

Storage requirements are calculated based on data point size of approximately 200 bytes including tags and metadata. With a data rate of approximately 1.4 kilobytes per second for seven devices publishing every 5 seconds, the system generates approximately 120 megabytes of data per day. Retention policies are configurable, with a default of 30 days for testing purposes, though this can be adjusted based on storage capacity and analysis requirements.

Database optimization strategies include time-based retention policies that automatically remove old data, downsampling techniques for long-term storage that reduce data granularity while maintaining important trends, and efficient indexing on device_id tags to enable fast queries and filtering operations.

6. Experiments and Results

6.1 System Integration Testing

End-to-end data flow testing verified the complete data pipeline from Wokwi simulation to Grafana visualization. The test procedure involved starting all services, running the Wokwi simulation, and monitoring data flow through each system component. Results demonstrated successful data transmission, storage, and visualization with latency measurements showing less than 2 seconds from sensor reading to dashboard update, meeting real-time monitoring requirements.

MQTT communication testing evaluated message delivery and reliability under various conditions. The procedure involved monitoring MQTT topics and verifying message delivery with QoS level 1. Results showed 100 percent message delivery rate with no observed message loss, indicating reliable communication infrastructure. Throughput testing confirmed the system handles seven devices publishing every 5 seconds without performance issues, with capacity for additional devices if needed.

Database performance evaluation measured InfluxDB write and query performance under typical operating conditions. Write latency measurements showed less than 10 milliseconds per data point, ensuring efficient data ingestion. Query latency for 24-hour historical queries remained under 100 milliseconds, providing responsive data retrieval for dashboard updates

and forecasting operations. Storage efficiency was demonstrated through compressed time-series data storage, optimizing disk space utilization.

6.2 Forecasting Accuracy

The forecasting accuracy experiment utilized 24 hours of simulated consumption patterns as training data, with a 6-hour forecast horizon for testing. Evaluation metrics included Mean Absolute Error and Mean Absolute Percentage Error to assess prediction quality.

Results demonstrated Mean Absolute Error values ranging from 15 to 25 watts, which is acceptable for the 0 to 500 watt measurement range, representing approximately 3 to 5 percent of the maximum value. Mean Absolute Percentage Error values ranged from 5 to 10 percent, indicating good accuracy for energy forecasting applications. Confidence intervals with 80 percent prediction intervals effectively captured actual values, demonstrating appropriate uncertainty quantification. Pattern recognition capabilities showed that Prophet successfully identified daily consumption patterns, enabling accurate predictions based on historical behavior.

Observations revealed that forecasts were more accurate for devices with consistent usage patterns, as expected for time-series forecasting models. Hardware-controlled devices showed more predictable behavior than random simulations, which aligns with real-world scenarios where actual appliances exhibit more consistent patterns than purely random data. Confidence intervals appropriately widened for longer forecast horizons, reflecting increased uncertainty for predictions further into the future.

6.3 Edge AI Control Performance

Predictive shutdown testing evaluated the system's ability to prevent overloads before they occur. Test scenarios involved situations where forecasts indicated devices would exceed the 400-watt threshold. Results showed the system automatically sent OFF commands 1 to 2 minutes before predicted overloads, providing sufficient time for device response. Effectiveness measurements showed 100 percent prevention rate in test scenarios, with false positive rates below 5 percent, indicating minimal unnecessary shutdowns while maintaining protection capabilities.

Multi-device coordination testing assessed the system's ability to handle multiple simultaneous alerts. Test scenarios involved multiple devices forecasted to exceed thresholds simultaneously. Results demonstrated the system's capability to handle multiple alerts independently, with response times under 5 seconds for processing and command dispatch, ensuring timely protection across all devices.

Manual override testing evaluated user control capabilities and system responsiveness. Test scenarios involved users manually controlling devices through the Grafana interface. Results

showed commands executed immediately with responsive interface feedback, and the AI control system properly respected manual overrides, ensuring user authority while maintaining automated protection capabilities.

6.4 System Scalability

Load testing evaluated system performance under various device counts and publishing rates. Testing with the current 7 devices and simulation up to 20 devices showed the system handles the current configuration comfortably with room for expansion. Node-RED processing utilized less than 1 percent CPU usage, indicating efficient resource utilization. InfluxDB demonstrated capability to handle over 100 writes per second, far exceeding current requirements. MQTT broker performance showed no bottlenecks, with capacity for additional devices and higher publishing rates.

Resource usage measurements showed total memory consumption of approximately 500 megabytes across all containers, average CPU utilization below 10 percent, and minimal network bandwidth usage of approximately 10 kilobytes per second. These measurements indicate efficient resource utilization with significant capacity for system expansion.

6.5 Dashboard Visualization

User interface evaluation assessed the dashboard's effectiveness in presenting information and enabling user interaction. Real-time updates refresh the dashboard every 5 seconds, providing current information without excessive resource consumption. Visual clarity was achieved through well-designed gauges and graphs that clearly display consumption patterns. Forecast visualizations effectively show prediction uncertainty through confidence bands, helping users understand forecast reliability. The control interface provides intuitive manual control through clearly labeled buttons.

User feedback indicated that the visualizations clearly display energy consumption trends, forecast graphs help users understand future consumption patterns, alert notifications provide timely warnings, and manual control capabilities give users final decision authority, balancing automation with user control.

6.6 Limitations and Challenges

Identified limitations include the simulation versus reality gap, where simulated data may not reflect real-world consumption patterns accurately, potentially affecting forecast accuracy and system behavior. Forecast accuracy depends on data quality and pattern consistency, with less predictable devices showing reduced prediction quality. Network dependency requires stable MQTT broker connection, with system functionality dependent on communication reliability. Scalability testing has not been performed beyond 20 devices, though theoretical limits appear higher based on resource measurements.

Challenges overcome during development included successfully implementing TLS configuration for secure MQTT communication in Wokwi MicroPython, standardizing JSON data format across all components for consistency, ensuring time synchronization across services for accurate temporal analysis, and implementing robust error handling and reconnection logic for system reliability.

7. Conclusion

The Smart Home Energy Optimization Assistant represents a significant step towards smarter, more efficient homes. By enabling and combining localized monitoring, forecasted predictions, and scheduled recommendations, the project allows users to gain control of their household energy consumption while reducing environmental impact.

Key achievements include successfully implementing an end-to-end IoT system from device simulation to visualization, demonstrating effective use of edge computing for real-time control decisions, integrating AI forecasting through Prophet for predictive energy management, creating user-friendly dashboards for monitoring and control, and achieving reproducible, containerized deployment that simplifies system setup and maintenance.

Technical contributions include a hybrid edge-cloud architecture that balances responsiveness and computational power, a predictive protection system that prevents overloads before they occur, scalable containerized infrastructure for easy deployment, and successful integration of multiple open-source technologies including MQTT, InfluxDB, Grafana, and Prophet.

Practical impact includes providing actionable insights into energy consumption patterns, enabling proactive energy management through forecasting, reducing risk of overloads through automated protection, and empowering users with manual control capabilities that maintain user authority while benefiting from automation.

Future enhancements could include implementing full AI-driven control using smart plug commands based on forecasts and user preferences, comparing energy consumption patterns across multiple households for benchmarking, integrating solar panel energy inputs to optimize usage schedules based on renewable energy availability, experimenting with LSTM models for potentially better forecast accuracy, developing mobile applications for remote monitoring and control, adding electricity pricing data to optimize usage during low-cost periods, and implementing reinforcement learning to adapt to user preferences automatically.

Reproducibility is ensured through Docker Compose configuration enabling one-command deployment, version-controlled code and configurations, clear documentation and setup instructions, and standardized data formats and protocols. This project demonstrates the feasibility and value of integrating IoT, edge computing, and artificial intelligence for smart energy management, providing a foundation for future smart home energy optimization systems.

8. References

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